ACE Cyber Project Charter: Fall 2025

1. Student(s) & Team Information

Principal Investigator (PI) Name(s): Teaundra Thompson
University: Charleston Southern University

2. Faculty / DoD Mentor Information

Faculty Advisor: Prof. Julie Henderson
DoD/ACE Cyber Mentor (if applicable): Prof. Bill Littleton

3. Project Title

Provide a clear, concise title for your project

_IDS Resiliency Against Evasion Attacks____

4. Problem Statement

Which research project did you select? Why is this important?

I selected the research project, AI IDS Resilience to Adversarial Evasion. This topic is important because government entities utilize IDSs. These systems are critical in cyber security infrastructure, and it is vital to assess the vulnerabilities of the AI within the systems.

5. Research Question

Define the primary question your project seeks to answer

What is the percentage drop of detection accuracy of the baseline IDS when faced with evasion attacks?

6. Hypothesis (if applicable)

What do you expect to discover or prove?

The IDS will detect large volume attacks with a high true positive rate, but when faced with evasive small volume attacks the true positive rate will decrease.

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7. Project Objectives

List 3–5 clear objectives for your project — measurable and achievable in ~3 months

1. Quantify accuracy deterioration under poisoning attacks

2. Quantify the success rate of adversarial attacks

3. Define how the IDS can be hardened against these attacks

8. Research Method Selection

Circle or highlight one and explain why you chose it:
- Applied Experiment
- Applied Observational Study

I chose applied experiment because I will need to create an IDS and test it in a closed environment. I will also be manipulating variables to receive necessary data.

9. Scope & Deliverables

What will you produce? What's in/out of scope?
Deliverables: IDS baseline model, Adversarial testing report, Solutions for model improvements, and adversarial code examples
Out-of-Scope Items: Novel poisoning techniques, and deployable resilient IDS

10. Resources & Tools Needed

List technologies, datasets, labs, or access required

- Lab environment that is supportive of ML development and has GPU resources
- ML datasets
- Adversarial ML frameworks

11. Timeline (3-Month Scope)

Month 1: Literature review and hypothesis
Month 2: Create adversarial examples and test them against IDS
Month 3: Consolidate findings and create presentation

12. Success Criteria & Metrics

How will success be measured? What are the KPIs or evaluation criteria?

Success will be evaluated and measured in these four aspects:

- IDS model detection accuracy
- Percentage drop in detection accuracy resulting from adversarial evasion
- False negative rate
- Number of adversarial techniques applied and tested successfully

13. Risks & Mitigations

Identify key risks — technical, data, schedule — and how you plan to address them

I foresee a possible risk with the technicalities of creating a baseline IDS and within the time constraint. I plan to address this risk by creating a consistent schedule and plan for completing these parts of the project. There's also a risk of the IDS not having enough resources to be trained thoroughly, but this risk can be mitigated by creating a decision tree model which is less resource intensive. Another risk is the availability of robust datasets. A lack of quality data to train the model will result in faulty simulation that does not reflect a realistic system under attack.

14. Expected Impact

How does this contribute to cybersecurity research for DoD?

This project contributes to DoD research by testing a prototype of a essential system for cyber security. This project will exemplify the vulnerabilities of these AI-based Intrusion Detection Systems and provide ideas on how to mitigate these vulnerabilities from advanced evasion techniques.